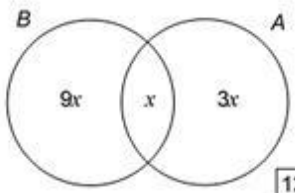


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates $P(\text{studies Physics}) \times P(\text{studies Geography})$ or Calculates $P(\text{studies Geography} \text{studies Physics})$ or $P(\text{studies Physics} \text{studies Geography})$	AO3.1b	M1	$P(P) \times P(G) = \frac{12}{24} \times \frac{8}{24} = \frac{1}{6}$
	Shows $P(\text{studies Physics}) \times P(\text{studies Geography}) = P(\text{studies Physics} \cap \text{studies Geography})$ and correctly concludes that the events are independent or Shows that the appropriate conditional probability is equal to $P(\text{studies Geography})$ or $P(\text{studies Physics})$ and correctly concludes that the events are independent	AO2.1	R1	$P(P \cap G) = \frac{4}{24} = \frac{1}{6}$ Hence $P(P) \times P(G) = P(P \cap G)$ Therefore events are independent
(b)	Uses conditional probability to calculate $P(M \cap B)$	AO3.1b	M1	$P(M \cap B) = P(M) \times P(B M)$
	Obtains the correct value of $P(M \cap B)$	AO1.1b	A1	$= \frac{1}{5} \times \frac{3}{8} = \frac{3}{40}$
	Uses the addition rule to calculate $P(M \cup B)$	AO1.1a	M1	$P(M) + P(B) - P(M \cap B)$
	Obtains the correct value of $P(M \cup B)$	AO1.1b	A1	$= \frac{1}{5} + \frac{1}{6} - \frac{3}{40}$ $= \frac{7}{24}$
				Total 6 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses conditional probability, either ① or ②	AO3.1b	M1	$\frac{P(A \cap B)}{P(A)} = \frac{1}{4}$ $\Rightarrow P(A) = 4P(A \cap B)$
Obtains both equations ① and ② correctly	AO1.1b	A1	$\frac{P(A \cap B)}{P(B)} = \frac{1}{10}$ $\Rightarrow P(B) = 10P(A \cap B)$
Evaluates $P(A \cup B)$ correctly PI	AO1.1b	B1	$P(A \cup B) = 1 - \frac{122}{200} = \frac{39}{100}$
Uses addition law	AO1.1a	M1	$P(A) + P(B) - P(A \cap B) = \frac{39}{100}$
Combines the three equations	AO1.1a	M1	$4P(A \cap B) + 10P(A \cap B) - P(A \cap B) = \frac{39}{100}$
Obtains correct probability, as a fraction or decimal	AO2.2b	A1	$P(A \cap B) = \frac{3}{100}$
ALT			
Produces a relevant Venn diagram	AO3.1b	M1	OR 200 
Labels Venn diagram correctly	AO1.1b	A1	
Forms correct equation to find x PI	AO1.1b	B1	$9x + x + 3x = 200 - 122$
Combines terms	AO1.1a	M1	$13x = 78$
Solves equation	AO1.1a	M1	$x = 6$
Obtains correct probability	AO2.2b	A1	$P(A \cap B) = \frac{6}{200} \text{ or } 0.03$
			Total 6 marks

Q3.

- (a) (i) $P(F \text{ \& } C) = \underline{0.3 \text{ or } 3/10 \text{ or } 30\%}$

Ratios (eg 3:10) are only penalised by 1 accuracy mark at first correct answer

CAO (0.3)

B1
(1)

(ii) $P(G \text{ or } S) = \underline{0.45 \text{ or } 45 / 100 \text{ or } 45\%}$
CAO (0.45)

B1
(1)

(iii) $P(C | F) = \frac{0.3 \text{ or } (i)}{0.55} =$

M1

$\underline{30 / 55 \text{ or } 6 / 11}$
CAO (6 / 11)

or

$\underline{(0.545 \text{ to } 0.55) \text{ or } (54.5\% \text{ to } 55\%)}$
AWFW (0.54545)

A1
(2)

(iv) $P(F | C') = \frac{0.25 \text{ or } (0.60 - 0.35)}{0.60}$
Correct numerator

M1

Correct denominator

M1

$\underline{25 / 60 \text{ or } 5 / 12}$
CAO (5 / 12)

or

$\underline{(0.416 \text{ to } 0.42) \text{ or } (41.6\% \text{ to } 42\%)}$
AWRT (0.41667)

A1
(3)

7

(b) $P = [P(F \& C)]^2 \text{ or } [P(F \& G)]^2$
Attempt at sum of at least 2 squared terms; $0 < \text{term} < 1$; not $(a + b)^2$

*May be implied by a **correct** expression or a **correct** answer*

M1

$$0.30^2 + 0.25^2 \text{ or } 0.09 + 0.0625 =$$

OE

Ignore additional terms or integer multipliers

*May be implied by a **correct** answer*

A1

$$\underline{1525 / 10000 \text{ or } 305 / 2000 \text{ or } 61 / 400}$$

CAO

or

$$(0.1525)$$

$$\underline{(0.152 \text{ to } 0.153) \text{ or } (15.2\% \text{ to } 15.3\%)}$$

AWFW

A1

3

[10]

Q4.

(a) (i) $P(F \text{ \& } C) = \underline{0.3 \text{ or } 3 / 10 \text{ or } 30\%}$

Ratios (eg 3:10) are only penalised by 1 accuracy mark at first correct answer

CAO

$$(0.3)$$

B1

(1)

(ii) $P(G \text{ or } S) = \underline{0.45 \text{ or } 45 / 100 \text{ or } 45\%}$

CAO

$$(0.45)$$

B1

(1)

(iii) $P(C | F) = \frac{0.3 \text{ or (i)}}{0.55} =$

M1

$$\underline{30 / 55 \text{ or } 6 / 11}$$

CAO

$$(6 / 11)$$

or

$$\underline{(0.54 \text{ to } 0.55) \text{ or } (54\% \text{ to } 55\%)}$$

AWFW

$$(0.54545)$$

A1
(2)

$$(iv) \quad P(R' | D) = \frac{0.25 \text{ or } (0.30 - 0.05)}{0.30}$$

Correct numerator

M1

Correct denominator

M1

$$\frac{25}{30} \text{ or } \frac{5}{6}$$

CAO (5 / 6)

or

$$\frac{(0.83 \text{ to } 0.834) \text{ or } (83\% \text{ to } 83.4\%)}{AWFW \quad (0.83333)}$$

A1
(3)

$$(v) \quad P(F | C') = \frac{0.25 \text{ or } (0.60 - 0.35)}{0.60}$$

Correct expression

M1

$$\frac{25}{60} \text{ or } \frac{5}{12}$$

CAO (5 / 12)

or

$$\frac{(0.416 \text{ to } 0.42) \text{ or } (41.6\% \text{ to } 42\%)}{AWRT \quad (0.41667)}$$

A1
(2,3)

9

(b) $P = [P(F \& C)]^2 + [P(F \& G)]^2$

*Attempt at **sum of at least 2 squared terms**; $0 < \text{term} < 1$; not $(a + b)^2$*
*May be implied by a **correct** expression or a **correct** answer*

M1

$$0.30^2 + 0.25^2 \quad \text{or} \quad 0.09 + 0.0625 =$$

OE

Ignore additional terms or integer multipliers
*May be implied by a **correct** answer*

A1

1525 / 10000 or 305 / 2000 or 61 / 400

CAO

or

(0.1525)

(0.152 to 0.153) or (15.2% to 15.3%)

AWFW

A1

3

[12]

Q5.

(a) (i) $P(A = 2) = 0.90 \times 0.95 = \underline{0.85 \text{ to } 0.86}$
AWFW (0.855 or 171 / 200 OE)

B1

(ii) $P(A = 1) = (0.90 \times 0.05) + (0.10 \times 0.95)$
May be implied by a **correct** answer

or $= 1 - [0.855 + (0.10 \times 0.05)]$
Do **not** ignore extra terms

M1

= 0.14
CAO (7 / 50 OE)

A1

3

(b) (i) $P(A_W \cap D_W) = 0.90 \times 0.80$
May be implied by a **correct** answer

M1

= 0.72
CAO (18 / 25 OE)

A1

2

$$(ii) \quad P(A_B \cap D_B) = (b)(i) \times 0.95 (\times 1)$$

$$\text{or} = 0.90 \times 0.80 \times 0.95 (\times 1)$$

or= (a)(i) × 0.80

May be implied by a **correct** answer

M1

0.68 to 0.685

AWFW (0.684 or 171 / 250 OE)

A1

2

(iii) $P(A_T \cap D'_T) = 0.95 \times 0 =$ 0
 CAO; award on value only

B1

1

(iv) $P(\text{neither}) = P([A'_W \cap D'_W] \cap [A'_T \cap D'_T])$
 $(1 - 0.90) \times (1 - 0.15)$
 Accept 0.085 or 17 / 200 OE
 Award M1 and m1 on value(s) only

M1

$(1 - 0.95) \times (1 - 0)$
 Accept 0.05 or 1 / 20 OE

m1

or
 $P(\text{neither}) =$
 $P(A'_W \cap A'_T) \cap P(D'_W | A'_W) \cap P(D'_T | A'_T)$

$(1 - 0.90) \times (1 - 0.95)$
 Accept 0.005 or 1 / 200 OE
 Award M1 and m1 on value(s) only

(M1)

$(1 - 0.15) \times (1 - 0)$
 Accept 0.85 or 17 / 20 OE

(m1)

$= 0.085 \times 0.05$ or 0.005×0.85
 OE

= 0.0042 to 0.0043
 AWFW (0.00425 or 17 / 4000 OE)

A1

3

[11]

Q6.

(a)

A	B		
----- T(0.9)	----- E(0.15)	0.135	
	----- T(0.75)	0.675	
	----- L(0.10)	0.090	
----- L(0.1)	----- T(0.35)	0.035	
	----- L(0.65)	0.065	

Correct shape

B1

Correct labels

B1

Correct probabilities

B1

3

(b) (i) $P(E \cup T @ B) = 0.9 \times 0.9 + 0.1 \times 0.35$
 $1 - (0.09 + 0.065)$

$= \underline{\underline{0.84 \text{ to } 0.85}}$
AWFW (0.845)

M1

A1

2

(ii) $P(T @ A | T @ B) =$

$\frac{0.9 \times 0.75}{(0.9 \times 0.75 + 0.1 \times 0.35)}$

$P(A | B)$ *used in (ii) or (iii)*

M1

$a \div (a + b)$ *with at least a correct*

m1

$= \frac{0.675}{0.71} = \underline{\underline{0.95 \text{ to } 0.951}}$
AWFW (0.95070)

A1

3

(iii) $P(L @ A | L' @ B) = \frac{0.1 \times 0.35}{(i)}$
F on (i)

AF1

$$= \frac{0.035}{0.845} = \underline{0.04 \text{ to } 0.042}$$

AWFW (0.04142)

A1

2

(c) $P((T @ A | L @ B) \cap (T' @ A | L @ B))$

$$\frac{0.9 \times 0.1}{1 - 0.845} \times \frac{0.1 \times 0.65}{1 - 0.845} \times 2$$

First expression (18 / 31)

M1

Second expression (13 / 31)

M1

$\times 2$

M1

$$= \underline{0.486 \text{ to } 0.49}$$

AWFW (0.48699)

A1

4

[14]

Q7.

- (a) (i) Table Method
(2- way with either R or C totals)

	A	A'	Total
E	0.55	0.05	0.60
E'	0.30		0.40
Total	0.85	0.15	1.00

0.15 or 0.4; CAO; allow fractions

B1

0.05 and 0.3; CAO; allow fractions

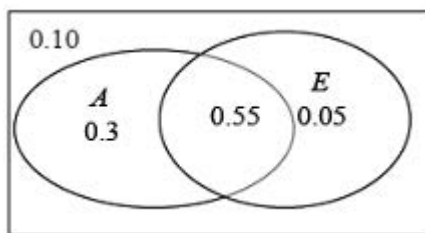
B1

0.1; AG so dependent on B1 B1

Bdep 1

3

Alternative solution 1
Venn Diagram Method



0.55; CAO

(B1)

0.3 and 0.05; CAO

(B1)

0.1; AG so dependent on B1 B1

(Bdep1)

(3)

Alternative solution 2

Formula Method

$$P(\geq 1) = 0.85 + 0.60 - 0.55$$

OR

$$0.85 + 0.60 - 0.55 + p = 1$$

OR

$$0.15 + 0.40 - 0.45$$

Full justification for numerical expression

(M2)

Insufficient justification or numerical expression only

(M1)

$$P(0) = 1 - P(\geq 1)$$

$$\text{OR } 1 - 0.9 = 0.1$$

$$0.9 + p = 1$$

$$\text{OR } 0.1$$

AG; gained from M2 or M1

(A1)

(3)

(ii) $P(\geq 1) = 0.9$ or $9/10$

CAO

B1

1

(iii) $P(1) = 0.3 + 0.05 = 1 - (0.55 + 0.10) = 0.35$ or $35/100$ or $7/20$

CAO

B1

1

- (b) (i) $P(3) = 0.55 \times 0.30$
OE; implied by correct answer

B1

$= 0.165$ or $165/1000$ or $33/200$
 CAO

B1

2

- (ii) $0.55 \times (1 - 0.3)$ or 0.385

M1

(0.3×0.75) or 0.225
 or (0.05×0.75) or 0.0375
 or (0.35×0.75) or 0.2625
At least one of these expressions or values

M1

$(0.385 + 0.2625) + 0.165$
OE; implied by correct answer

B1

$= 0.812$ to 0.813
 AFWW (0.8125)

or $\frac{8125}{10000}$ or $\frac{1625}{2000}$ or $\frac{325}{400}$ or $\frac{65}{80}$ or $\frac{13}{16}$

CAO

A1

©2025 Exam Papers Practice. All rights reserved.

4

Alternative solution 1

$0.1 \times (1 - 0.4)$ or 0.06

(M1)

(0.3×0.25) or 0.075
 or (0.05×0.25) or 0.0125
 or (0.35×0.25) or 0.0875
 or (0.1×0.4) or 0.04
At least one of these expressions or values

(M1)

$1 - (0.1875)$
OE; implied by correct answer

(B1)

$$= 0.812 \text{ to } 0.813$$

AWFW (0.8125)

CAO for equivalent fraction

(A1)

(4)

Alternative solution 2

$$(0.55 + p) \quad \text{where } 0 < p < 0.45$$

(M1)

$$(0.3 \times 0.75) \text{ or } 0.225$$

$$\text{or } (0.05 \times 0.75) \text{ or } 0.0375$$

$$\text{or } (0.35 \times 0.75) \text{ or } 0.2625$$

At least one of these expressions or values

(M1)

$$0.55 + 0.2625$$

OE; implied by correct answer

(B1)

$$= 0.812 \text{ to } 0.813$$

AWFW (0.8125)

CAO for equivalent fraction

(A1)

(4)

[11]

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q8.

*Ratios (eg 194:640) are only penalised by
 1 accuracy mark at first correct answer*

(a) (i) $P(B = 3) = \underline{194 / 640 \text{ or } 97 / 320 \text{ or } 0.303 \text{ or } 30.3\%}$

CAO or AWRT (0.303125)

B1

1

(ii) $P(T \geq 2) = \frac{172 + 256 + 135}{640} \text{ or } 1 - \frac{77}{640} \text{ or } \frac{563}{640} = \underline{563 / 640}$
CAO

M1

or (0.879 to 0.88) or (87.9% to 88%)

AWFW (0.879688)

A1

2

$$(iii) \quad P(B = 3 \text{ \& } T > 2) = \frac{72+99+16}{640} \text{ or } \frac{194-7}{640} \text{ or } \frac{187}{640}$$

M1

$$= \frac{187}{640} \text{ or } 0.292 \text{ or } 29.2\%$$

CAO or AWRT (0.292188)

A1

2

$$(iv) \quad P(B < 3 \mid T = 2) = \frac{(14+67+72)}{172} \text{ or } \frac{172-19}{172} \text{ or } \frac{153}{172}$$

Correct numerator
Correct denominator
(accept both ÷ 640)

M1

M1

$$= \frac{153}{172}$$

CAO

A1

$$\text{or } (0.888 \text{ to } 0.89) \text{ or } (88.8\% \text{ to } 89\%)$$

AWFW (0.889535)

3

$$(b) \quad P(2T \cap 3T \cap \geq 4T \mid B = 3) = \frac{72}{194} \times \frac{99}{193} \times \frac{16}{192}$$

Correct 3 values multiplied in numerator
Correct 3 values multiplied in denominator
 $0.371 \times 0.513 \times 0.083$ (all AWRT)
 $\Rightarrow$ M1 M1 (OE products)

M1

M1

abc multiplied by 6 or 3

$$0 < (a, b \text{ \& } c) < 1$$

M1

$$= 0.095 \text{ to } 0.0952$$

$$\text{AWFW } (0.095187)$$

A1

Notes 1 Incorrect answer with no working $\Rightarrow$ 0 marks

$$= \frac{153}{172}$$

CAO

or (0.888 to 0.89) or (88.8% to 89%)
 AFWW (0.889535)

A1

Answers as fractions, percentages or ratios lose accuracy (A & B) marks in (b) & (c)

3

(b) (a)(i) $\times$ (a)(ii) $\neq$ (a)(iii)
Attempted

M1

since

$$0.303 \times 0.88 = \underline{0.265 \text{ to } 0.27 \neq 0.292}$$

AWFW & AWRT

A1

2

SC Any **correct fully-explained** reasoning, using other than answers from part (a), which results in an inequality ($\neq$) **with both sides as numerically correct decimals** (to 3 dp) $\Rightarrow$ B1 (eg $P(B = 3) = 0.303 \neq P(B = 3 | T = 2) = 72 / 172 = 0.419$) but no / unclear / incomplete reasoning or no / incorrect / incomplete numerical work $\Rightarrow$ B0

(c) $P(2T \cap 3T \cap \geq 4T | B = 3) = \frac{72}{194} \times \frac{99}{193} \times \frac{16}{192}$

Correct 3 values multiplied in numerator

Correct 3 values multiplied in denominator

M1

$$0.371 \times 0.513 \times 0.083 \text{ (all AWRT)}$$

$\Rightarrow$ M1 M1 (OE products)

M1

abc multiplied by 6 or 3

$$0 < (a, b \text{ \& } c) < 1$$

M1

$$= \underline{0.095 \text{ to } 0.0952}$$

AWFW

$$(0.095187)$$

A1

4

Notes 1 Incorrect answer with no working $\Rightarrow$ 0 marks

4 A denominator of ${}^{194}C_3 = 1198144 \Rightarrow$ M2 (2nd & 3rd M1 marks)

$$(e) \quad P(S | O) = \frac{\frac{0.15 \times (1 - 0.10)}{1 - (iii)}}{0.9025} = \frac{0.135}{0.9025}$$

Numerator

M1

Denominator

M1

$$= 0.149 \text{ to } 0.15$$

AWFW (0.14958)

A1

3

[10]

Q11.

$$(a) \quad (i) \quad P(M \cap C) = \frac{175}{645} = \frac{35}{129} = 0.271$$

AWRT; accept either correct fraction

B1

1

$$(ii) \quad P(M) = \frac{519}{645} = \frac{173}{215} = 0.804 \text{ to } 0.805$$

AWFW; accept either correct fraction

B1

1

$$(iii) \quad P(LD) = \frac{63}{645} = \frac{21}{215} = 0.097 \text{ to } 0.098$$

AWFW; accept either correct fraction

B1

1

$$(iv) \quad P(L | F) = \frac{94}{126} = \frac{47}{63}$$

$$\text{Accept } \frac{94}{645} \div \frac{126}{645}$$

M1

$$= 0.746$$

A1

AWRT

2

$$\begin{aligned}
 \text{(b) } P(L \cap L | F) &= \left(\frac{94}{126} \times \frac{93}{125} \right) \text{ or } \frac{8742}{15750} \\
 \text{Or } \left(\frac{47}{63} \times \frac{93}{125} \right) &\text{ or } \frac{4371}{7875} \text{ or } \frac{1457}{2625}
 \end{aligned}$$

B1

$$= 0.555$$

B1

AWRT

2

$$\text{(c) } P(L \cap C \cap (LD + 0))$$

$$= \frac{349}{645} \times \frac{193}{644} \times \frac{63 + 40}{643}$$

Correct numerator

M1

Correct denominator

M1

SC The three correct fractions identified but not multiplied $\Rightarrow$ M1 M0 M0 A0

$\times 6$ or 3

EXAM PAPERS PRACTICE
 Note that a denominator of $\begin{pmatrix} 645 \\ 3 \end{pmatrix}$

M1

$$= 0.155 \text{ to } 0.157$$

$\Rightarrow$ M2 (second and third M1 marks)

AWFW

A1

4

NB: 0.026 with no working $\Rightarrow$ M1 only

0.026 $\times$ 6 = 0.156 with no working $\Rightarrow$ 4 marks

[11]

Q12.

(a) (i) $P(M \cap C) = \frac{175}{645} = \frac{35}{129} = 0.271$

AWRT; accept either correct fraction

B1

1

(ii) $P(M) = \frac{519}{645} = \frac{173}{215} = 0.804$ to 0.805

AWFW; accept either correct fraction

B1

1

(iii) $P(LD) = \frac{63}{645} = \frac{21}{215} = 0.097$ to 0.098

AWFW; accept either correct fraction

B1

1

(iv) $P(L|F) = \frac{94}{126} = \frac{47}{63}$
Accept $\frac{94}{645} \div \frac{126}{645}$

$= 0.746$

AWRT

M1

EXAM PAPERS PRACTICE

A1

©2025 Exam Papers Practice. All rights reserved.

2

(v) $P(M|L') = \frac{519 - 255}{645 - 349} = \frac{175 + 54 + 35}{193 + 63 + 40}$

Allow one arithmetic slip

M1

Allow one arithmetic slip

M1

$= \frac{264}{296} = \frac{132}{148} = \frac{66}{74} = \frac{33}{37}$

Any of these implies M1 M1

$= 0.891$ to 0.893

AWFW

A1

3

$$\begin{aligned}
 \text{(b)} \quad P(L \cap L \mid F) &= \left(\frac{94}{126} \times \frac{93}{125} \right) \text{ or } \frac{8742}{15750} \\
 &\text{Or } \left(\frac{47}{63} \times \frac{93}{125} \right) \text{ or } \frac{4371}{7875} \text{ or } \frac{1457}{2625}
 \end{aligned}$$

B1

$$= 0.555$$

AWRT

B1

2

$$\text{(c)} \quad P(L \cap C \cap (LD + O))$$

$$= \frac{349}{645} \times \frac{193}{644} \times \frac{63 + 40}{643}$$

Correct numerator

Correct denominator

M1

M1

SC The three correct fractions identified but not multiplied $\Rightarrow$ M1 M0 M0 A0

$\times 6$ or 3

EXAM PAPERS PRACTICE

Note that a denominator of $\begin{pmatrix} 645 \\ 3 \end{pmatrix}$

$\Rightarrow$ M2 (second and third M1 marks)

M1

$$= 0.155 \text{ to } 0.157$$

AWFW

A1

4

NB: 0.026 with no working $\Rightarrow$ M1 only

$0.026 \times 6 = 0.156$ with no working $\Rightarrow$ 4 marks

[14]

Q13.

$$\text{(a)} \quad R \sim (15, 0.45)$$

- (i) $P(R \leq 5) = 0.26$ (0) to 0.261
AWFW (0.2608)

B1

1

- (ii) $P(R > 10) = 1 - P(R \leq 10)$
Requires '1 -'
Accept 3dp rounding or truncation
 $= 1 - (0.9745 \text{ or } 0.9231)$
Can be implied by 0.025 to 0.026 but not by 0.0769 to 0.077

M1

$$= 0.025 \text{ to } 0.026$$

AWFW (0.0255)

A1

2

- (iii) $P(R = 6) = 0.4522 - (a)(i) \text{ or } = \binom{15}{6} (0.45)^6 (0.55)^9$
Can be implied by a correct answer

M1

$$= 0.191 \text{ to } 0.192$$

AWFW (0.1914)

A1

2

- (iv) $P(5 \leq R \leq 10) = 0.9745 \text{ or } 0.9231 (p_1)$
Accept 3dp rounding or truncation
 $p_2 - p_1 \Rightarrow M0 M0 A0$
 $(1 - p_2) - p_1 \Rightarrow M0 M0 A0$
 $p_1 - (1 - p_2) \Rightarrow M1 M0 A0$
only providing result > 0

M1

Minus 0.1204 or 0.2608(p_2)
Accept 3dp rounding or truncation

M1

$$= 0.853 \text{ to } 0.855$$

AWFW (0.8541)

A1

Or

B (15, 0.45) terms stated for at least 3 values within
 $4R \leq 11$ gives probability

Can be implied by a correct answer

(M1)

= 0.853 to 0.855

AWFW (0.8541)

(A2)

- (b) (i) $P(S) = 0.85$ plus 1 minus
CAO; requires 'plus' or 'minus'

B1

$(0.15 \times 0.80) \quad (0.15 \times 0.20)$
CAO; not simply 0.12 or 0.03

B1

= 0.97

AG

2

NB: $(0.85 \times 0.20) + 0.80 \Rightarrow$ B0 B0

$(0.85 \times 0.20) + (0.85 \times 0.80) + (0.15 \times 0.80) \Rightarrow$ B0 B1

- (ii) $P(S \geq 48) = 0.81$ to 0.82 or 0.5553 or 0.9372

Accept 3dp rounding or truncation

*M2 for the three correctly expressed terms
 for B (50, 0.03) or B (50, 0.97) added*

M2

= 0.81(0) to 0.811

AWFW (0.8108)

A1

3

NB: Answer = 0.4447 or 0.1892 or 0.0628 $\Rightarrow$ M1 only

- (iii) $p = 1 - 0.85 = 0.15$

*CAO; may be implied by correct answer
 or correct expression for mean*

B1

Mean, $\mu = 80 \times 0.15 = 12$

CAO

B1

2

SC Mean = 9.6 $\Rightarrow$ B1 only

[15]

Q14.

(a)

	<i>J</i>	<i>J'</i>	Total
<i>W</i>	0.55	0.10	0.65
<i>W'</i>	0.15	0.20	0.35
Total	0.70	0.30	1.00

0.35 and 0.7; CAO

B1

0.55; CAO

B1

0.1 and 0.2; CAO

B1

Notes:

Use of Venn or tree diagrams **without** table completion $\Rightarrow$ B0 B0 B0

Printed table not completed but constructed and completed on a different page $\Rightarrow$ B1 B1 B1 max

Accept fractional answers

Do not accept percentages

©2025 Exam Papers Practice. All rights reserved.

3

(b) P(purchases exactly one)

$$= P(W \cap J') + 0.15$$

*Only c's equivalent to 0.10 **shown and added to** 0.15*

*Can be implied by **correct** answer*

M1

$$= 0.10 + 0.15$$

$$= \mathbf{0.25 \text{ or } 25/100 \text{ or } 5/20 \text{ or } 1/4}$$

CAO

A1

2

(c) (i) $P(W \cup J) = \mathbf{0.8} \neq P(W) + P(J) = \mathbf{1.35}$

or $P(W \cap J) = 0.55 (>0)$; accept if indicated in a Venn diagram

or $P(W) + P(J) = 1.35 > 1$ or impossible

Any one of these three **seen**

Ignore contradictions, explanations & justifications

B1

(ii) $P(W | J) = 0.55/0.70 = 0.79$

&/# $p(W) = 0.65$

Do **not** accept use of W' and/or J'

AWRT

B1

or $P(J | W) = 0.55/0.65 = 0.85$

&/# $P(J) = 0.70$

or $P(W) \times P(J) = 0.45$ to 0.46

&/# $P(W \cap J) = 0.55$

Any one of these three **seen**

Ignore contradictions, explanations & justifications

AWFW

Bdep1

3

[8]

Q15.

(a) (i)

	J	J'	Total
W	0.55	0.10	0.65
W'	0.15	0.20	0.35
Total	0.70	0.30	1.00

B1

0.35 and 0.7; CAO

B1

0.55; CAO

B1

0.1 and 0.2; CAO

Notes:

Use of Venn or tree diagrams **without** table completion $\Rightarrow$ B0 B0 B0

Accept fractional answers

Do not accept percentages

Printed table not completed but constructed and completed on a different page $\Rightarrow$ B1 B1 B1 max

3

(ii) P(purchases exactly one)

$$= P(W \cap J') + 0.15$$

$$= 0.10 + 0.15$$

*Only c's equivalent to 0.10 **shown and added to 0.15***

*Can be implied by **correct** answer*

M1

$$= 0.25 \text{ or } 25/100 \text{ or } 5/20 \text{ or } 1/4$$

CAO

A1

2

(iii) (A) $P(W \cup J) = 0.8 \neq P(W) + P(J) = 1.35$

or $P(W \cap J) = 0.55 (>0)$; accept if indicated in a Venn diagram

or $P(W) + P(J) = 1.35 > 1$ or impossible

*Any one of these three **seen***

Ignore contradictions, explanations & justifications

B1

(B) $P(W | J) = 0.55/0.70 = 0.79$

*Do **not** accept use of W' and/or J'*

AWRT 25 Exam Papers Practice. All rights reserved.

B1

$$\neq p(W) = 0.65$$

Bdep1

$$\text{or } P(J | W) = 0.55/0.65 = 0.85$$

$$\neq P(J) = 0.70$$

$$\text{or } P(W) \times P(J) = 0.45 \text{ to } 0.46$$

$$\neq P(W \cap J) = 0.55$$

*Any one of these three **seen***

Ignore contradictions, explanations & justifications

AWFW

(b) **Do not allow multiplying factors in (b)**

(i) $P(0) = 0.15 \times 0.40 \times 0.45$

*Can be implied by **correct** answer or
 $1 - (0.2265 + 0.466 + 0.2805)$*

B1

= 0.027 or 27/1000
 CAO

B1

2

(ii) $P(2) = 0.85 \times 0.60 \times 0.45 = 0.2295$
 $+ 0.85 \times 0.40 \times 0.55 = 0.1870$
 $+ 0.15 \times 0.60 \times 0.55 = 0.0495$

For either method:

At least two bold expressions correct

Only one bold expression correct

*Can be implied by **correct** answer*

M2
 (M1)

or

= 1 – (0.027 + 0.2265 + 0.2805)

For second method:

Must have '1 –' for any marks

= 0.466 or 466/1000 or 233/500

CAO; **do not** imply this from (i)

A1

3

[13]

Q16.

(a) (i) $P(G) = 0.15$
 CAO

B1

1

(ii) $P(A \cap \leq 1) = 0.60 \times 0.55 = 0.33$
 CAO

B1

1

(iii) $P(\leq 24) = (0.60 \times 0.80) + (0.25 \times 0.85) + (0.15 \times 0.75)$
May be implied

M1

$= 0.48 + 0.2125 + 0.1125 = \mathbf{0.805}$
 CAO

A1

2

(iv) $P(B | \leq 24) = \frac{P(B \cap \leq 24)}{P(\leq 24)}$
Used; may be implied

M1

$= \frac{0.25 \times 0.85}{\text{(iii)}} = \frac{0.2125}{0.805}$
F on (iii)

AF1

$= \mathbf{0.264}$
 AWR T

A1

3

(b) (i) $P(3 @ B | \leq 24) = [(a)(iv)]^3$
Used; may be implied

M1

$= \mathbf{0.018 \text{ to } 0.0185}$
 AFWW (0.01839)

A1

2

(ii) $P(\text{same station} | \leq 24) = [P(A | \leq 24)]^3 + (b)(i) + [P(G | \leq 24)]^3$
Used; may be implied

M1

$= \left(\frac{0.48}{0.805} \right)^3 + (b)(i) + \left(\frac{0.1125}{0.805} \right)^3$

At least 1 term correct; allow (b)(i) providing it is a (cond prob)³

M1

All 3 terms correct

M1

$$= 0.2120 + 0.0184 + 0.0027 = \mathbf{0.233}$$

AWRT (0.23312)

A1

4

[13]

Q17.

**Ratios (eg 65:1000) are only penalised
by 1 mark at first correct answer**

- (a) (i) $P(\text{all 3 walk}) = 0.65 \times 0.40 \times 0.25$
*Can be implied by **correct** answer*

M1

$$= 65/1000 = 13/200 = 0.065$$

CAO; do not confuse with 0.65

A1

2

- (ii) $P(\text{Rita by bus}) = 0.25 \times (1 - 0.15) \times (1 - 0.20)$
*Can be implied by **correct** answer*

M1

$$= 17/100 = 0.17$$

CAO

A1

2

- (iii) $P(2 \text{ cycle})$
 $= 0.10 \times 0.45 \times (0.25 + 0.20)$
 $= 0.02025$
 $+ 0.10 \times (0.40 + 0.15) \times 0.55$
 $= 0.03025$
 $+ (0.65 + 0.25) \times 0.45 \times 0.55$
 $= 0.22275$
 (0.27325)

*CAO at least 1 of these 3 terms or
equivalent but allow a '× 3'*

B1

$$P(3 \text{ cycle}) = 0.10 \times 0.45 \times 0.55$$

$$= 0.02475$$

CAO

B1

$$P(\geq 2 \text{ cycle}) = P(2 \text{ cycle}) + P(3 \text{ cycle})$$

Sum of 4 or 7 terms each a product of 3 probabilities but not '× 3'

M1

$$= 0.298$$

CAO

A1

or

$$P(0 \text{ cycle}) = 0.90 \times 0.55 \times 0.45 = 0.22275$$

CAO

(B1)

$$\begin{aligned}
 &P(1 \text{ cycles}) \\
 &= 0.10 \times 0.55 \times 0.45 = 0.02475 \\
 &+ 0.90 \times 0.45 \times 0.45 = 0.18225 \\
 &(0.47925)
 \end{aligned}$$

CAO at least 1 of these 3 terms but allow a '× 3'

(B1)

$$\begin{aligned}
 &+ 0.90 \times 0.55 \times 0.55 = 0.27225 \\
 &P(\geq 2 \text{ cycle}) \\
 &= 1 - [P(0 \text{ cycle}) + P(1 \text{ cycles})] \\
 &1 - [\text{sum of 4 terms each a product of 3 probabilities but not '× 3'}]
 \end{aligned}$$

(M1)

$$1 - 0.702 = 0.298$$

CAO

(A1)

4

(b) (i) $P(WW) = (0.65 \times 0.90) = 0.585$
 $P(CC) = (0.10 \times 0.70) = 0.070$
 CAO either

B1

$$P(WW \text{ or } CC) = 0.585 + 0.070$$

Sum of 2 terms each a product of 2 probabilities

M1

$$= 0.655$$

CAO; or equivalent

A1

3

(ii) $P(\text{different}) = 1 - (b)(i) = 0.345$

F on (b)(i) providing $0 < p < 1$

B1F

1

[12]

Q18.

Ratios (eg 63:100) are only penalised by 1 mark at first correct answer F marks can only be awarded if $0 < p < 1$

(a) $P(J) = 0.9 \quad P(R | J) = 0.7 \quad P(R | J') = 0.2$

(i) $P(\text{both at trough}) = 0.9 \times 0.7$

*Can be implied by **correct** answer*

M1

$= 0.63 = 63/100$

CAO

A1

2

(ii) $P(\text{neither at trough}) = (1 - 0.9) \times (1 - 0.2)$
 $= 0.1 \times 0.8$

*Can be implied by **correct** answer*

M1

$= 0.08 = 8/100 = 4/50 = 2/25$

CAO

A1

2

(iii) $P(\text{at least one at trough}) = (1 - (ii))$
 $= 0.92 = 92/100 = 46/50 = 23/25$

*F on (ii) or **correct** answer*

B1F

1

(b) (i)

<i>M</i>	<i>M'</i>	Total
----------	-----------	-------

<i>D</i>	0.40	0.35	0.75
<i>D'</i>	0.20	0.05	0.25
Total	0.60	0.40	1.00

Both row and column totals
ie 0.25 and 0.40; CAO

B1

Three table values
ie 0.35 and 0.20 and 0.05; CAO

B1

Notes:

Use of Venn or tree diagrams **without** table completion $\Rightarrow$ B0 B0

2

(ii) **Accept answers $\div$ 1.00**

(A) $P(\text{neither at gate}) = 0.05$
F on table
or **correct** answer by 'otherwise'

B1F

1

(B) $P(\text{only Daisy at gate}) = 0.35$
F on table
or **correct** answer by 'otherwise'

B1F

1

(C) $P(\text{exactly one at gate}) = P(D \cap M') + P(D' \cap M)$
Only **correct two** values from c's table
shown and added
Can be implied by **correct** answer

M1

$0.35 + 0.20 = 0.55$
F on table
or **correct** answer by 'otherwise'

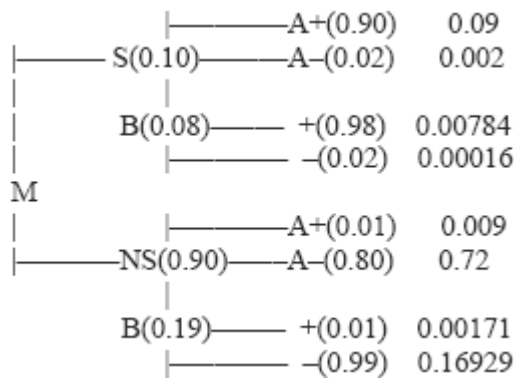
A1F

2

[11]

Q19.

(a)



S and NS with Ps or %s

B1

2 × (A+ and A−) with Ps or %s

B1

2 × (B) with Ps or %s

B1

2 × (B+ and B−) with Ps or %s

B1

Basic shape with labels, but without Ps or %s

(B2, 1)

Note:

The following BF and AF marks are dependent on an essentially correctly-shaped tree diagram

4

- (b) (i) **(A)** $P(S \text{ and } -) = 0.002 + 0.00016 = \mathbf{0.00216}$
F on (a); otherwise CAO

B1F

- (B)** $P(NS \text{ and } +) = 0.009 + 0.00171 = \mathbf{0.01071}$
F on (a); otherwise AWRT 0.0107

B1F

2

- (ii) $E(N) = 10000 \times [(A) + (B)]$
Or equivalent

M1

$$= 128.6 \text{ to } 128.7 \Rightarrow \mathbf{130}$$

CAO

A1F

2

(c) (i)
$$P(S | +) = \frac{P(S \text{ and } +)}{P(+)} =$$

Used

M1

$$\frac{0.09 + 0.00784}{0.09 + 0.00784 + 0.009 + 0.00171} = \frac{0.09784}{0.10855}$$

F on (a)
Otherwise correct

A1F

= 0.901 to 0.902
AWRT (0.90134)

A1

(ii)
$$P(NS | -) = \frac{P(NS \text{ and } -)}{P(-)} =$$

Used; only if not scored in (i)

(M1)

$$\frac{0.72 + 0.16929}{0.002 + 0.00016 + 0.72 + 0.16929} = \frac{0.88929}{0.89145}$$

F on (a) and/or denominator (c)(i)
Otherwise correct

A1F

= 0.997 to 0.998
AWFW (0.99758)

A1

Special cases:

Only numerators correct $\Rightarrow$ (M1) B1 B1

Only denominators correct $\Rightarrow$ (M1) B1 B1

5

[13]

Q20.

(a)

In (a), ratios (eg 100:160) are only penalised by 1 mark at first correct answer

(i) $P(P) = 100/160 = 50/80 = 25/40 = 10/16 = 5/8 = 0.625$
 CAO

B1

1

(ii) $P(S') = 1 - \frac{32}{160}$ **or** $P(S') = \frac{32}{160}$

Or equivalent

Ignore labels of S' & S

*Can be implied by **correct** answer*

M1

$= 128/160 = 64/80 = 32/40 = 16/20 = 8/10 = 4/5 = 0.8$
 CAO

A1

2

(iii) $P(S \text{ or } H) = P(S \cup H) =$
 $\frac{60 + 32 - 18}{160}$ **or** $\frac{60 + 14}{160}$ **or** $\frac{32 + 8 + 16 + 18}{160}$

Or equivalent

*Can be implied by **correct** answer*

M1

$= 74/160 = 37/80 = 0.462 \text{ to } 0.463$
 CAO/AWFW (0.4625)

A1

2

EXAM PAPERS PRACTICE

© Exam Papers Practice. All rights reserved.

(iv) $P(T | P) = \frac{\frac{30}{160}}{(i)}$

Or equivalent

*Can be implied by **correct** answer*

but watch for $\frac{18}{160}$ or $\frac{48}{160}$

M1

$= 3/100 = 3/10 = 0.3$
 CAO

A1

2

(b) $P(1C \text{ \& } 1R \text{ \& } 1S) =$

$$\frac{24}{160} \times \frac{56}{159} \times \frac{32}{158}$$

Multiplication of any 3 different given subject totals

M1

Multiplication of 160, 159 & 158

M1

$$(0.15 \times 0.35220 \times 0.20253)$$

$\times 6$

Accept 3 dp accuracy

Award for $3 \leq \text{multiplier} \leq 6$

M1

$$= 0.064 \text{ to } 0.0644$$

AWFW (0.0642)

Do not accept a fraction as answer

*A **correct** answer can imply 4 marks*

A1

Special Case:

(Any given subject total) + 160 seen anywhere in (b)

Can award if no marks scored in (b)

Accept a decimal equivalent

(M1)

4

[11]

Q21.

(a) (i) $P(B \& B) = (0.30 \times 0.80) + (0.55 \times 0.10) + (0.15 \times 0.30)$

*Use of **3 possibilities** each the product of **2 probabilities***

M1

$$= 0.24 + 0.055 + 0.045 = 0.34$$

CAO; AG

A1

2

(ii) $P(HB \cap \text{Coastal}) = 0.55 \times 0.65$

*Can be implied by **correct** answer*

M1

$$= 143/400 \text{ or } 0.357 \text{ to } 0.358$$

CAO/AWFW (0.3575)

A1

2

$$(iii) \quad P(\text{Coastal} | \text{HB}) = \frac{P(\text{coastal} \cap \text{HB})}{P(\text{HB})}$$

$$\frac{\text{answer to (ii)}}{\sum (3 \times 2) \text{ probabilities}}$$

M1M1

$$= \frac{0.3575}{(0.3 \times 0.15) + (0.3575) + (0.15 \times 0.5)}$$

For (ii)

A1F

$$= \frac{0.3575}{0.4775} = 143/191 \text{ or } 0.747 \text{ to } 0.75$$

CAO/AWFW (0.74869)

A1

4

$$(b) \quad P(\text{City} | \text{HB}) = \frac{0.3 \times 0.15}{P(\text{HB})} = \frac{0.045}{0.4775} = \frac{90}{955}$$

M1

$$P(\text{Country} | \text{HB}) = \frac{0.15 \times 0.5}{P(\text{HB})} = \frac{0.075}{0.4775} = \frac{30}{191}$$

$$\text{Or } \left(1 - (a)(iii) - \frac{0.045}{0.4775} \right)$$

M1

Thus Probability =

$$\frac{0.045}{P(\text{HB})} \times \frac{0.3575}{P(\text{HB})} \times \frac{0.075}{P(\text{HB})}$$

Multiplication of 3 different probabilities

M1

Multiplied by 3! = 6

CAO

B1

$$\begin{aligned}
 &= 0.09424 \times 0.74869 \times 0.15707 \times 6 \\
 &= 0.063 \text{ to } 0.068 \\
 &\quad \text{AWFW (0.06649)}
 \end{aligned}$$

A1

5

[13]

Q22.

(a) (i) $P(D \cap A) = 0.85 \times 0.80$
Can be implied by correct answer

M1

$$\begin{aligned}
 &= 0.68 \\
 &\quad \text{CAO; OE}
 \end{aligned}$$

A1

2

(ii) $P(D' \cap A) = 0.15 \times 0.30$
Can be implied by correct answer

M1

$$\begin{aligned}
 &= 0.045 \\
 &\quad \text{CAO; OE}
 \end{aligned}$$

A1

2

(iii) $P(\text{One}) = (ii) + (0.85 \times 0.20)$
Can be implied by correct answer

M1

$$\begin{aligned}
 &= 0.045 + 0.17 = 0.215 \\
 &\quad \text{CAO; OE}
 \end{aligned}$$

A1

2

(b) $P(D' \cap A') =$
 0.15×0.7
or $= 0.105$
CAO; OE; must be stated somewhere

B1

$$1 - (0.68 + 0.215)$$

$$P((D \cap A)^3 \cap (D' \cap A')) =$$

$$[(a)(i)]^3 \text{ or } 0.68^3$$

M1

$$\times 0.105 \times 4$$

Either; with ≥ 1 ($\times$ sign)

M1

$$= 0.132$$

AWRT (0.13206)

A1

4

[10]

Q23.

(a) (i) $P(G') = 1 - 0.70 = 0.3(0)$

CAO; OE

B1

1

(ii) $P(G \cap S') = 0.70 - (0.25 \text{ or } 0.55 \text{ or } 0.45)$
 or $1 - 0.55$

Can be implied only if answer is correct

M1

$$= 0.45$$

CAO; OE

A1

2

(iii) $P(1 \text{ only}) = 0.70 + 0.55 - (2 \times 0.25)$

or $1 - 0.25$

or $0.45 + 0.30$

*Can be implied only if answer is correct;
 allow no ($\times 2$) but not by implication from answer*

M1

$$= 0.75$$

CAO; OE

A1

2

(b) $P(G' \cap G' \cap G' \cap G') = [(a)(i)]^4$

Can be implied by correct answer
Ignore multiplier of 4

M1

$$= 0.0081$$

CAO; OE

A1

2

(c) $P(H_G) = P(A_G \cap H_G) + P(A_{G'} \cap H_G) =$
 $(0.70 \times 0.60) \text{ or } 0.42$

Can be implied by correct answer
Ignore additional terms

M1

$$(0.30 \times 0.10) \text{ or } 0.03$$

Can be implied by correct answer

M1

$$= 0.42 + 0.03 = 0.45$$

CAO; OE

A1

3

(d) $P(H_O) = 1 - [0.35 + (c)]$

Can be implied by correct answer

M1

$$= 0.2(0)$$

CAO; OE

A1

2

[12]

Q24.

(a) $P(\text{Blue}) = \frac{160}{400} = 0.4 \text{ or } \frac{2}{5} \text{ or } \frac{160}{400}$

CAO; or equivalent

B1

In (b) to (e), method marks are for single fractions, or equivalents, only

1

(b) $P(\text{Marker}) = \frac{280}{400}$
 $270 \leq \text{Numerator} \leq 290$ and
 $\text{Numerator} < \text{Denominator} \leq 400$

M1

$= 0.7$ or $\frac{7}{10}$ or $\frac{280}{400}$
 CAO; or equivalent

A1

2

(c) $P(B \text{ or } M) = P(B \cup M) =$
 $\frac{160 + 280 - 119}{400} = \frac{280 + 41}{400} = \frac{321}{400}$
 $290 \leq \text{Numerator} \leq 321$ and
 $\text{Numerator} < \text{Denominator} \leq 400$

M1

$= 0.802$ to 0.803 or $\frac{321}{400}$
 AWW/CAO (0.8025)

A1

2

(d) $P(\text{Green} \mid \text{Highlighter}) = P(G \mid H) = \frac{42}{120}$
 $\text{Numerator} = 42$ and
 $110 \leq \text{Denominator} \leq 120$

M1

$= 0.35$ or $\frac{7}{20}$ or $\frac{42}{120}$
 CAO; or equivalent

A1

2

(e) $P(\text{Non-Permanent} \mid \text{Red}) = P(P' \mid R) = \frac{21}{90}$
 $\text{Numerator} = 21$ and
 $80 \leq \text{Denominator} \leq 90$

M1

$$= 0.233 \text{ to } 0.234 \text{ or } \frac{7}{30} \text{ or } \frac{21}{90}$$

AWFW/CAO (0.2333)

A1

2

[9]

Q25.

(a)

	— E(0.30)	0.0750
— B(0.25)	— M(0.50)	0.1250
	— W(0.20)	0.0500
	— E(0.40)	0.0240
— S(0.60)	— M(0.45)	0.2700
	— W(0.15)	0.0900
	— E(0.55)	0.0825
— D(0.15)	— M(0.35)	0.0525
	— W(0.10)	0.0150

B, S & D with 3 probabilities

B1

3 × (E, M & W) each with 3 probabilities

B2

≥ 1 × (E, M & W) (each) with 3 probabilities

(B1)

3

(b) (i) $P(E) = (0.25 \times 0.3) + (0.6 \times 0.4) + (0.15 \times 0.55)$
 $= 0.075 + 0.24 + 0.0825 =$
 $\geq 1 \text{ term correct}$

M1

0.397 to 0.398 or 159/400
 AFWW/CAO (0.3975)

A1

2

(ii) $P(D | E) = \frac{0.0825}{(b)(1)} =$
Or equivalent

M1

0.207 to 0.208 or 11/53
AWFW/CAO (0.2075)

A1

2

- (c) $X \sim B(10, (b) (ii))$
Used

M1

$$P(X = 4) = \binom{10}{4} (0.2075)^4 (0.7925)^6 =$$

ft on (b)(ii)

A1ft

0.0955 to 0.0975
AWFW (0.09645)

A1

3

[10]

Q26.

Ratios: Penalise first occurrence only of a correct answer

(a) (i) $P(\text{English}) = \frac{14 + 8}{50} =$

Correct expression; PI

B1

$$\frac{22}{50} \text{ or } \frac{11}{25} \text{ or } 0.44$$

CAO; OE

B1

2

(ii) $P(\text{Irish} | \text{back}) = \frac{P(\text{Irish} \cap \text{back})}{P(\text{back})} = \frac{8}{\sum(\text{back})} =$

Used; may be implied by values or answer

M1

$$\frac{6}{23} \text{ or } 0.26 \text{ to } 0.261$$

CAO/AWFW (6/50 $\Rightarrow$ 0)

A1

2

$$(iii) \quad P(\text{forward} | \text{not Scottish}) = \frac{P(\text{forward} \cap \text{not Scottish})}{P(\text{not Scottish})} = \frac{14 + 5 + 6}{50 - 4} = \frac{27 - 2}{50 - 4} =$$

Used; OE
May be implied by values or answer

M1

$$\frac{25}{46} \text{ or } 0.54 \text{ to } 0.544$$

CAO/AWFW (25/50 $\Rightarrow$ 0)

A1

2

$$(b) \quad P(4 \times \text{English}) = \left(\frac{22}{50}\right) \times \left(\frac{21}{49}\right) \times \left(\frac{20}{48}\right) \times \left(\frac{19}{47}\right) =$$

Reducing non-tabulated value 4 times

M1

Reducing 50 and multiplying 4 terms
(ignore multipliers)

M1

$$\frac{175560}{5527200} \text{ or } \frac{209}{6580}$$

or 0.0317 to 0.032

CAO/AWFW

A1

3

[9]

Q27.

Ratios: Penalise first occurrence only of a correct answer

$$(a) \quad (i) \quad P(\text{Welsh back}) = \frac{7}{50} \text{ or } 0.14$$

CAO; OE

B1

1

(ii) $P(\text{English}) = \frac{14 + 8}{50} =$
Correct expression; PI

B1

$\frac{22}{50}$ or $\frac{11}{25}$ or 0.44
CAO; OE

B1

2

(iii) $P(\text{not English}) = 1 - (\text{ii}) =$
 $\frac{28}{50}$ or $\frac{14}{25}$ or 0.56
ft on (ii) if used; $0 < p < 1$

B1ft

1

(iv) $P(\text{Irish} | \text{back}) = \frac{P(\text{Irish} \cap \text{back})}{P(\text{back})} = \frac{6}{\sum(\text{back})} =$
Used; may be implied by values or answer

M1

$\frac{6}{23}$ or 0.26 to 0.261
CAO/AWFW ($6/50 \Rightarrow 0$)

A1

2

(v) $P(\text{forward} | \text{not Scottish}) = \frac{P(\text{forward} \cap \text{not Scottish})}{P(\text{not Scottish})} =$
Used; OE
May be implied by values or answer

M1

$\frac{14 + 5 + 6}{50 - 4} = \frac{27 - 2}{50 - 4} =$
 $\frac{25}{46}$ or 0.54 to 0.544

CAO/AWW (25/50 $\Rightarrow$ 0)

A1

2

(b) $P(4 \times \text{English}) =$

$$\left(\frac{22}{50}\right) \times \left(\frac{21}{49}\right) \times \left(\frac{20}{48}\right) \times \left(\frac{19}{47}\right) =$$

Reducing non-tabulated value 4 times

M1

*Reducing 50 and multiplying 4 terms
(ignore multipliers)*

M1

$$\frac{175580}{5527200} \text{ or } \frac{209}{6580}$$

or 0.0317 to 0.032

CAO / AWW

A1

3

[11]

Q28.

(a) $P(\geq 18 \mid \text{Road}) = 0.85$

CAO; OE; **not** 85

B1

1

(b) $P(18 \text{ to } 64) =$

$P(\text{Route}) \times P(18 \text{ to } 64 \mid \text{Route}) =$

*Use of 3 possibilities, each the product
of 2 probabilities*

M1

$$(0.25 \times 0.80) + (0.60 \times 0.35) + (0.55 \times 0.40)$$

At least 1 term correct

A1

$$= 0.20 + 0.21 + 0.22 = 0.63$$

CAO; OE

A1

3

(c) $P(\text{FR} \cap > 64) = P(\text{FR}) \times P(> 64 \mid \text{FR})$
 $= 0.35 \times 0.15$

Correct expression

B1

$= 0.052 \text{ to } 0.053$

AWFW (0.0525)

B1

2

(d) $P(\text{FR} \mid > 64) = \frac{(c)}{P(> 64)} =$

$$\frac{\text{answer (c)}}{\sum (3 \times 2) \text{ probabilities}}$$

M1M1

$$\frac{0.0525}{(0.25 \times 0.05) + (0.35 \times 0.15) + (0.40 \times 0.35)}$$

At least 2 terms correct

A1

$$= \frac{0.0525}{0.0125 + 0.0525 + 0.1400} = \frac{0.0525}{0.205}$$

CAO

A1

0.256 or $\frac{21}{82}$

AWRT/CAO; OE

A1

5

[11]

Q29.

(a) $P(X) = 0.3 \quad P(Y) = 0.4 \quad P(Z) = 0.2$

(i) $P(X \cap Y \cap Z) = 0.3 \times 0.4 \times 0.2 = 0.024$

M1

1

(ii) $P(X' \cap Y' \cap Z') = 0.7 \times 0.6 \times 0.8$

At least 2 correct terms

M1

$$= 0.0336$$

CAO

A1

2

(iii) $P(X' \cap Y' \cap Z) = 0.7 \times 0.6 \times 0.2$

Correct numerical expression

M1

$$= 0.084$$

CAO

A1

2

(b) $P(W | Z) = 0.9 \quad P(W | Z') = 0.25$

(i) $P(Z \cap W) = 0.2 \times 0.9$

Correct numerical expression

M1

$$= 0.18$$

CAO

A1

2

(ii) $P((Z \cap W) \cup (Z' \cap W))$

or

$$1 - [P((Z \cap W) \cup (Z' \cap W'))]$$

$$= 0.2 \times (1 - 0.9)$$

M1

+

$$0.2 \times 0.9 \quad \text{or} \quad (b) (i)$$

$$(1 - 0.2) \times 0.25$$

$$(1 - 0.2) \times (1 - 0.25)$$

Cannot score an M1 in both methods

M1

$$= 0.02 + 0.20$$

$$1 - (0.18 + 0.60)$$

$$= 0.22$$

CAO

A1

3

[10]

Q30.

	O (R)	1 (S)	2 (T)	≥ 3	T
D (D)	24	32	41	23	120
SD (D')	40	37	88	35	200
T	64	69	129	58	320

(a) (i) $P(D) = \frac{120}{320}$ or $\frac{3}{8}$ or 0.375

CAO; or equivalent

B1

1

(ii) $P(D \cap R) = \frac{24}{320}$ or $\frac{3}{40}$ or 0.075

CSO; or equivalent

B1

1

(iii) $P(D | R) = \frac{P(D \cap R)}{P(R)} = \frac{(ii)}{P(R)} = \frac{\frac{24}{320}}{\frac{64}{320}}$

M0 if independence assumed

M1

$$= \frac{24}{64} \text{ or } \frac{3}{8} \text{ or } 0.375$$

CAO; or equivalent

A1

2

$$(iv) \quad P(R | D') = \frac{P(R \cap D')}{P(D')} = \frac{\frac{40}{200}}{\frac{200}{320}}$$

numerator

allow independence assumed

denominator

M1 M1

$$= \frac{40}{200} \text{ or } \frac{1}{5} \text{ or } 0.2$$

CAO; or equivalent

A1

3

- (b) (i) R and S or R and T or S and T
not D and D'

EXAM PAPERS PRACTICE

- (ii) $P(D) = 0.375 = P(D | R)$ or (i) = (iii)

B1

1

M1

$$\begin{aligned} P(D) \times P(R) &= 0.375 \times 0.2 \\ &= 0.075 = P(D \cap R) \text{ or (ii)} \\ \text{or } P(R | D) &= P(R) = 0.2, \text{ etc} \end{aligned}$$

so YES

A1

2

- (c) (i) A semi-detached house
CAO

B1

or two children (or both)

or equivalent

B1

2

- (ii) A detached house and/or with less than two children

B1

CAO

(0 or 1 must not include 'both')

B1

2

[14]

Q31.

	O (R)	1 (S)	2 (T)	≥ 3	T
D (D)	24	32	41	23	120
SD (D')	40	37	88	35	200
T	64	69	129	58	320

(a) (i) $P(D) = \frac{120}{320}$ or $\frac{3}{8}$ or 0.375

EXAM PAPERS PRACTICE

CAO; or equivalent

B1

1

(ii) $P(D \cap R) = \frac{24}{320}$ or $\frac{3}{40}$ or 0.075

CSO; or equivalent

B1

1

(iii) $P(D \cup T) = \frac{120 + 88}{320} = \frac{129 + 24 + 32 + 23}{320}$

M1

$$= \frac{208}{320} \text{ or } \frac{13}{20} \text{ or } 0.65$$

CAO; or equivalent

A1

2

$$(iv) \quad P(D | R) = \frac{P(D \cap R)}{P(R)} = \frac{(ii)}{P(R)} = \frac{\cancel{24}/(320)}{\cancel{64}/(320)}$$

M0 if independence assumed

M1

$$= \frac{24}{64} \text{ or } \frac{3}{8} \text{ or } 0.375$$

CAO; or equivalent

A1

2

$$(v) \quad P(R | D') = \frac{P(R \cap D')}{P(D')} = \frac{\cancel{40}/(320)}{\cancel{200}/(320)}$$

numerator
 allow independence assumed
 denominator

M1M1

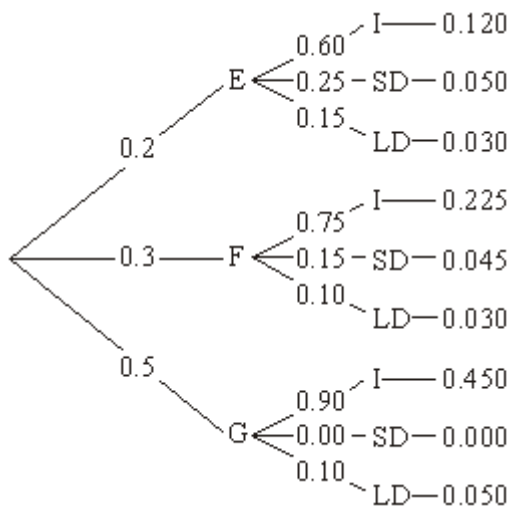
$$= \frac{40}{200} \text{ or } \frac{1}{5} \text{ or } 0.2$$

CAO; or equivalent

		A1	3
(b)	(i)	R and S or R and T or S and T <i>not D and D'</i>	
		B1	1
	(ii)	$P(D) = 0.375 = P(D R)$ or (i) = (iv) $P(D) \times P(R) = 0.375 \times 0.2$ $= 0.075 = P(D \cap R)$ or (ii) or $P(R D) = P(R) = 0.2$, etc	
		M1	
		so YES	
		A1	2
(c)	(i)	A semi-detached house CAO	
		B1	
		or two children (or both) <i>or equivalent</i>	
		B1	2
	(ii)	A detached house and/or with CAO	
		B1	
		less than two children <i>(0 or 1 must not include 'both')</i>	
		B1	2

[16]

Q32.



- (a) (i) $P(G \cap I) = 0.5 \times 0.9 = 0.45$
CAO; or equivalent

B1

1

- (ii) $P(I) = (i) + P(E \cap I) + P(F \cap I)$
3 possibilities

M1

3

$$= 0.45 + (0.2 \times 0.6) + (0.3 \times 0.75)$$

*≥ 1 correct **new** term*

A1

$$= 0.45 + 0.12 + 0.225 = 0.795$$

CAO; or equivalent

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

A1

3

$$(iii) \quad P(G | I) = \frac{P(G \cap I)}{P(I)}$$

Attempted use of Bayes' Theorem

M1

$$= \frac{(i)}{(ii)} = \frac{0.45}{0.795} = 0.566$$

m1

AWRT; or equivalent

A1

3

$$(b) \quad P(E | SD) = \frac{P\{E \cap SD\}}{P\{SD\}}$$

Correct use of Bayes' Theorem

M1

$$= \frac{0.2 \times 0.25}{(0.2 \times 0.25) + (0.3 \times 0.15)} =$$

Numerator (B1 if no Bayes' Theorem)

A1

$$\frac{0.05}{0.05 + 0.045}$$

Denominator (B1 if no Bayes' Theorem)

A1

$$= \frac{0.05}{0.095} = 0.526$$

Papers Practice. All rights reserved.

AWRT; or equivalent

A1

4

[11]

Q33.

(a) (i)

	<i>M</i>	<i>A</i>	<i>S</i>	<i>T</i>
<i>M</i>	38	369	303	710

<i>F</i>	26	275	643	944
<i>T</i>	64	644	946	1654

$$P(F) = 944/1654 (= 0.571)$$

Use of

M1

1

$$(ii) \quad P(F \cap A) = 275/1654 (= 0.166)$$

Use of

M1

1

$$(iii) \quad P(F \cup A) = \frac{944 + 369}{1654}$$

Use of; OE

M1

$$= 1313/1654 \text{ or } 0.793 \text{ to } 0.795$$

CAO/AFW (0.7938)

A1

2

$$(iv) \quad P(F | A) = \frac{\text{their (ii)}}{644/1654}$$

Use of

M1

$$= 275/644 \text{ or } 0.426 \text{ to } 0.428$$

CAO/AFW (0.4270)

A1

2

$$(b) \quad (i) \quad P(MMM) = \frac{710 \times 709 \times 708}{1654 \times 1653 \times 1652}$$

Use of (without replacement)

M1

$$= 0.0788 \text{ to } 0.0791$$

AWFW (no fraction) (0.07891)

A1

2

(ii) $P(MFF) = \frac{710 \times 944 \times 943 \times 3}{1654 \times 1653 \times 1652}$
Use of one combination of MFF (without replacement)

M1

Use of multiplier of 3

M1

$= 0.419 \text{ to } 0.421$
AWFW (no fraction) (0.4198)

A1

3

- (c) (i) Female (and) Academic
 CAO

B1

1

- (ii) Male

Not female $\Rightarrow$

B1

B0

OR

'OR' must be clearly stated or implied

Academic (or both)

B1

2

Addition of 'not both' $\Rightarrow$

B0

[14]

Q34.

$P(D) = 0.6$ (60%)
 (40%)

$P(D') = 0.4$

$P(d | D) = 0.75$

$P(d | D') = 0.05$

$P(i | D) = 0.15$

$P(i | D') = 0.15$

$P(n | D) = 0.10$

$P(n | D') = 0.80$

$D \Rightarrow$ Diseased fish

$d \Rightarrow$ diseased)

$i \Rightarrow$ inconclusive) test

$n \Rightarrow$ non-diseased)

(a) (i) $P(D \cap d)$
 $= P(D) \times P(d \perp D) = 0.6 \times 0.75$

M1

$= 0.45$

CAO (9/20)

A1

2

(ii) $P(d) = P(D \cap d) + P(D' \cap d)$
 $= (i) + P(D') \times P(d \mid D')$
(a)(i) + (1 prob)

M1

$= 0.45 + (1 - 0.6) \times 0.05$
ft on (a)(i)

m1ft

$= 0.45 + 0.02 = 0.47$
 CAO

A1

3

(iii) $P(\text{correct}) = P(D \cap d) + P(D' \cap d')$
 $= (i) + P(D') \times P(d' \mid D')$
Correct expression/values
ft on (a)(i)

M1ft

Or
 $= 0.45 + (1 - 0.6) \times 0.80$
 $= 0.45 + 0.32 = 0.77$
 CAO

A1

2

(b) $P([d \mid D] \cap [d \mid D] \cap [d' \mid D])$
 $= [P(d \mid D)]^2 \times P(d' \mid D)$
(Prob)² × (Prob); or equivalent

M1

multiplied by 3

Multiplier of 3

M1

$$= 0.75^2 \times 0.10 \times 3 = 0.168 \text{ to } 0.169$$

AWFW (0.16875)

A1

3

[10]

Q35.

$P(F) = 0.8$ $P(D | F) = 0.9$ $P(D | F') = 0.4$

(a) (i) $P(F \cap D)$
 $= P(F) \times P(D | F) = 0.8 \times 0.9$

M1

$$= 0.72$$

CAO (18/25)

A1

2

(ii) $P(F' \cap D')$
 $= P(F') \times P(D' | F') = (1 - 0.8) \times (1 - 0.4)$

M1

$$= 0.2 \times 0.6 = 0.12$$

CAO (3/25)

A1

2

(b) $P(M) = 0.7$

(i) $P(F \cap D \cap M) = P(F) \times P(D | F) \times P(M)$
(a)(i) $\times P(M)$, ignore multipliers

M1

$$= (a)(i) \times P(M) = 0.72 \times 0.7$$

Or equivalent; ft on (a)(i) < 1

A1ft

$$= 0.504$$

CAO (63/125)

A1

3

(ii) $P(2 \text{ in } 3)$
 $= P(F \cap D \cap M')$

$$+ P(F \cap D' \cap M)$$

$$+ P(F' \cap D \cap M)$$

At least 2 permutations of 3 events seen, or implied by multiplication of 3 correct probabilities at least twice

Ignore multipliers e.g. $\times 3$

M1

$$= 0.8 \times 0.9 \times 0.3$$

$$+ 0.8 \times 0.1 \times 0.7$$

A2

$$+ 0.2 \times 0.4 \times 0.7$$

At least 2 correct expressions
(Exactly 1 correct expression)

(A1)

$$= 0.216 + 0.056 + 0.056$$

$$= 0.328$$

CAO (41/125)

A1

4

[11]



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.